

Math 6601 Homework 8

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Problem 1.

(a) Let

$$A = \begin{bmatrix} 2 & 3 \\ -2 & -6 \\ 1 & 0 \end{bmatrix}.$$

Define $v_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 3 \\ -6 \\ 0 \end{bmatrix}$, the columns of A . Then set

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{3}v_1.$$

Using Gram-Schmidt orthogonalization, we get $q_2 = \frac{p_2}{\|p_2\|}$, where

$$p_2 = v_2 - \langle v_2, q_1 \rangle q_1 = v_2 - 2v_1 = \begin{bmatrix} -1 \\ -2 \\ -2 \end{bmatrix}.$$

Then $q_2 = \frac{1}{3}p_2$. To complete the Q matrix in the full QR factorization of A , we need a unit vector q_3 that is orthogonal to both q_1 and q_2 . Set $v_3 = \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix}$. Then $q_3 = \frac{p_3}{\|p_3\|}$ does the job, where

$$p_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2 = v_3 - v_1 + 2p_2 = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix}.$$

Then $q_3 = \frac{1}{6}p_3$. Thus, the Q matrix in the QR factorization is

$$Q = [q_1 \quad q_2 \quad q_3] = \frac{1}{3} \begin{bmatrix} 2 & -1 & -2 \\ -2 & -2 & -1 \\ 1 & -2 & 2 \end{bmatrix},$$

and the R factor is

$$R = \begin{bmatrix} \langle v_1, q_1 \rangle & \langle v_2, q_1 \rangle \\ 0 & \langle v_2, q_2 \rangle \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}.$$

(b) Let

$$A = \begin{bmatrix} -4 & -4 \\ -2 & 7 \\ 4 & -5 \end{bmatrix}.$$

Define $v_1 = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -4 \\ 7 \\ -5 \end{bmatrix}$, the columns of A . Then set

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{6}v_1.$$

Using Gram-Schmidt orthogonalization, we get $q_2 = \frac{p_2}{\|p_2\|}$, where

$$p_2 = v_2 - \langle v_2, q_1 \rangle q_1 = v_2 + \frac{1}{2}v_1 = \begin{bmatrix} -6 \\ 6 \\ -3 \end{bmatrix}.$$

Then $q_2 = \frac{1}{9}p_2$. To complete the Q matrix in the full QR factorization of A , we need a unit vector q_3 that is orthogonal to both q_1 and q_2 . Set $v_3 = \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix}$. Then $q_3 = \frac{p_3}{\|p_3\|}$ does the job, where

$$p_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2 = v_3 - v_1 + \frac{1}{3}p_2 = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}.$$

Then $q_3 = \frac{1}{6}p_3$. Thus, the Q matrix in the QR factorization is

$$Q = [q_1 \quad q_2 \quad q_3] = \frac{1}{3} \begin{bmatrix} -2 & -2 & 1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix},$$

and the R factor is

$$R = \begin{bmatrix} \langle v_1, q_1 \rangle & \langle v_2, q_1 \rangle \\ 0 & \langle v_2, q_2 \rangle \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ 0 & 9 \\ 0 & 0 \end{bmatrix}.$$

Problem 2.

- (a) To find the least-squares solution of $Ax = b$, we can equivalently find the least-squares solution of $Rx = Q^T b$, where $A = QR$ is a QR factorization of A . Using the factorization from 1. (a), this means we want to find the least-squares solution of

$$\begin{bmatrix} 3 & 6 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ -1 & -2 & -2 \\ -2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}.$$

Since the last equation does not depend on x , the residual comes only from the first two equations, and is therefore minimized when x solves the first two equations exactly. Hence, $x_2 = -1$ from the second equation, and $x_1 = 4$ from the first equation is the least-square solution of $Ax = b$.

- (b) To find the least-squares solution of $Ax = b$, we can equivalently find the least-squares solution of $Rx = Q^T b$, where $A = QR$ is a QR factorization of A . Using the factorization from 1. (b), this means we want to find the least-squares solution of

$$\begin{bmatrix} 6 & -3 \\ 0 & 9 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -2 & -1 & 2 \\ -2 & 2 & -1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 9 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ 7 \end{bmatrix}.$$

Since the last equation does not depend on x , the residual comes only from the first two equations, and is therefore minimized when x solves the first two equations exactly. Hence, $x_2 = \frac{4}{9}$ from the second equation, and $x_1 = -\frac{11}{18}$ from the first equation is the least-square solution of $Ax = b$.

Problem 3.

- (a) Let Q be an $n \times n$ matrix with columns $q_1, q_2, \dots, q_n$. Then Q is orthogonal if and only if the columns $q_1, q_2, \dots, q_n$ of Q are pairwise orthogonal unit vectors.

Proof. Suppose that Q is orthogonal. Then $Q^T Q = I$, that is,

$$I = Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ q_2^T q_1 & q_2^T q_2 & \cdots & q_2^T q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \cdots & q_n^T q_n \end{bmatrix}$$

Thus,

$$\langle q_i, q_j \rangle = q_i^T q_j = I_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

This implies that $q_1, q_2, \dots, q_n$ are pairwise orthogonal unit vectors, as $\|q_i\| = \sqrt{\langle q_i, q_i \rangle} = 1$.

Conversely, suppose that $q_1, q_2, \dots, q_n$ are pairwise orthogonal unit vectors. Then

$$q_i^T q_j = \langle q_i, q_j \rangle = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

It follows that

$$Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ q_2^T q_1 & q_2^T q_2 & \cdots & q_2^T q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \cdots & q_n^T q_n \end{bmatrix} = I.$$

Then Q is orthogonal. □

- (b) Let A and B be orthogonal $n \times n$ matrices. Then AB is orthogonal.

Proof. We have

$$(AB)^T (AB) = B^T A^T AB = B^T IB = B^T B = I$$

by the orthogonality of A and B . Thus, AB is orthogonal. □

Problem 4.

- (a) Consider one step of the Gram-Schmidt method for computing the QR factorization of a (nonsingular, so that all the steps get done) $m \times m$ matrix A with columns $v_1, v_2, \dots, v_m$:

$$p_k \leftarrow v_k - \sum_{i=1}^{k-1} \langle v_k, q_i \rangle q_i, \quad q_k \leftarrow \frac{p_k}{\|p_k\|},$$

which must be computed for $k = 1, 2, \dots, m$. Implementing these formulas would work roughly as follows. We count additions and subtractions (A/S) and multiplications and divisions (M/D).

1. Compute $\langle v_k, q_i \rangle$ (these are stored in the R matrix) for $i = 1, 2, \dots, k-1$. This requires $(k-1)(m-1)$ additions and $(k-1)m$ multiplications. A/S: $(k-1)(m-1)$, M/D: $(k-1)m$.
2. Compute $\langle v_k, q_i \rangle q_i$ and sum for $i = 1, 2, \dots, k-1$ and subtract from v_k . This requires $(k-1)m$ multiplications and $(k-1)m$ additions. A/S: $(k-1)m$, M/D: $(k-1)m$.
3. Compute $q_k = \frac{p_k}{\|p_k\|}$. Note that $\langle v_k, q_k \rangle = \|p_k\|$, so calculating the norm gives the diagonal of the R matrix without needing to compute the inner product (we have to calculate the norm first anyway). The norm requires m multiplications and $m-1$ additions and one square root, which we will exclude, followed by m divisions. A/S: $m-1$, M/D: $2m$.

In total, step k requires

$$\begin{aligned} \text{A/S: } & (k-1)(m-1) + (k-1)m + m-1 = 2km - m - k, \\ \text{M/D: } & (k-1)m + (k-1)m + 2m = 2km. \end{aligned}$$

Summing from $k = 1$ to m gives the total cost:

$$\begin{aligned} \text{A/S: } & \sum_{k=1}^m (2km - m - k) = m^2(m+1) - m^2 - \frac{m(m+1)}{2} = m^3 - \frac{1}{2}m^2 - \frac{1}{2}m, \\ \text{M/D: } & \sum_{k=1}^m 2km = m^2(m+1) = m^3 + m^2. \end{aligned}$$

- (b) Consider the k th step of computing the QR factorization of an $m \times m$ matrix A . We count the number of additions and subtractions (A/S) and multiplications and divisions (M/D) required only to obtain the R factor and store the reflection directions (this is the only way to get the $\frac{2}{3}m^3$ cost; computing Q makes this method more expensive).

1. We start with a $(m-k+1) \times (m-k+1)$ matrix A_k , and let u_k be the first column of A_k . To construct the Householder reflector $\hat{H}_k$, we first need to compute the reflection axis $p_k = \frac{u_k - v_k}{\|u_k - v_k\|}$,

where $v_k = \begin{bmatrix} \|u_k\| \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. This requires one subtraction for $u_k - v_k$, then two norms, requiring a total

of $2(m-k)$ additions and $2(m-k+1)$ multiplications (and 2 square roots, which we exclude). It also requires a vector-scalar division, which consists of $m-k+1$ individual divisions. A/S: $2(m-k)+1$, M/D: $3(m-k+1)$.

2. Next, we compute $R_k = H_k R_{k-1}$, where we define $R_0 = A$. The algorithm will terminate with $R = R_{m-1}$. Since $H_k = \begin{bmatrix} I_{k-1} & 0 \\ 0 & \hat{H}_k \end{bmatrix}$, the first $k-1$ rows of R_k are the same as R_{k-1} . Noting that R_{k-1} is upper triangular in the first $k-1$ columns, it follows that the first $k-1$ columns of R_k will also be the same as those of R_{k-1} , saving us some computation. Let $\hat{R}_{k-1}$ be the bottom $(m-k+1)$ rows and columns of R_{k-1} , and define $\hat{R}_k$ similarly. Then

$$\hat{R}_k = \hat{H}_k \hat{R}_{k-1} = (I - 2p_k p_k^T) \hat{R}_{k-1} = \hat{R}_{k-1} - (2p_k)(p_k^T \hat{R}_{k-1}).$$

Computing $p_k^T \hat{R}_{k-1}$ is a $1 \times (m-k+1)$ by $(m-k+1) \times (m-k+1)$ matrix multiplication, which requires $(m-k+1)^2$ multiplications and $(m-k+1)(m-k)$ additions. Then computing $2p_k$ times the result is a $(m-k+1) \times 1$ by $1 \times (m-k+1)$ matrix multiplication, which requires $(m-k+1)^2 + m-k+1$ multiplications (if we do $2p_k$ first) and zero additions. Lastly, subtracting $\hat{R}_{k-1} - 2p_k p_k^T \hat{R}_{k-1}$ requires $(m-k+1)^2$ additions. Thus we have A/S: $(m-k+1)^2 + (m-k+1)(m-k)$, M/D: $2(m-k+1)^2 + m-k+1$.

Summing the additions and subtractions from $k = 1$ to $m - 1$, we have

$$\begin{aligned} \text{A/S: } \sum_{k=1}^{m-1} 2(m-k) + 1 + (m-k+1)^2 + (m-k+1)(m-k) &= \frac{2}{3}m^3 + \mathcal{O}(m^2), \\ \text{M/D: } \sum_{k=1}^{m-1} 2(m-k+1)^2 + 4(m-k+1) &= \frac{2}{3}m^3 + \mathcal{O}(m^2). \end{aligned}$$

Problem 5.

Let $u, v \in \mathbf{R}^n$ be distinct, nonzero vectors of the same length. Then the Householder matrix $H = I - 2pp^T$ is orthogonal, where $p = \frac{u-v}{\|u-v\|}$.

Proof. We have

$$\begin{aligned} H^T H &= (I - 2pp^T)^T (I - 2pp^T) = (I^T - 2(pp^T)^T)(I - 2pp^T) \\ &= (I - 2(p^T)^T p^T)(I - 2pp^T) = (I - 2pp^T)(I - 2pp^T) \\ &= I - 4pp^T + 4pp^T pp^T = I \end{aligned}$$

because $p^T p = \frac{(u-v)^T(u-v)}{\|u-v\|^2} = 1$. This implies that H is orthogonal. $\square$

Problem 6.

(a) Consider A given in 1. (a). We want to find a QR factorization of A using Householder reflectors. We

want to construct a reflector taking the first column v_1 into $u_1 = \begin{bmatrix} \|v_1\| \\ 0 \\ 0 \end{bmatrix}$. To do this, we set

$$p_1 = \frac{u_1 - v_1}{\|u_1 - v_1\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}.$$

Then the reflector we want is

$$H_1 = I - 2p_1 p_1^T = I - \frac{1}{3} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ -2 & -1 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

Next, we compute $A_1 = H_1 A$:

$$A_1 = H_1 A = \begin{bmatrix} 3 & 6 \\ 0 & 0 \\ 0 & -3 \end{bmatrix}.$$

Now we focus on the submatrix excluding the top row and left column of A_1 , which is $\hat{A}_1 = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$. We

want to construct a reflector $\hat{H}_2$ taking $u_2 = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$ to $v_2 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. To do this, we set

$$p_2 = \frac{u_2 - v_2}{\|u_2 - v_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \end{bmatrix},$$

and

$$\widehat{H}_2 = I - 2p_2p_2^T = I - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Then set $H_2 = \begin{bmatrix} 1 & 0 \\ 0 & \widehat{H}_2 \end{bmatrix}$. Multiplying H_2A_1 gives the matrix R in the QR factorization of A :

$$H_2H_1A = H_2A_1 = R = \begin{bmatrix} 3 & 6 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}.$$

Inverting H_2H_1 , and using the fact that Householder reflectors are self-inverse, we obtain the Q factor from $A = H_1H_2R$, so that $Q = H_1H_2$, or

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ -2 & -1 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ -2 & -2 & 1 \\ 1 & -2 & -2 \end{bmatrix},$$

which is the same as the one obtained in 1. (a) up to the arbitrary sign of the third column of Q .

- (b) Consider A given in 1. (b). We want to find a QR factorization of A using Householder reflectors. We want to construct a reflector taking the first column v_1 into $u_1 = \begin{bmatrix} -\|v_1\| \\ 0 \\ 0 \end{bmatrix}$. To do this, we set

$$p_1 = \frac{u_1 - v_1}{\|u_1 - v_1\|} = \frac{1}{\sqrt{24}} \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}.$$

Then the reflector we want is

$$H_1 = I - 2p_1p_1^T = I - \frac{1}{12} \begin{bmatrix} 4 & -4 & 8 \\ -4 & 4 & -8 \\ 8 & -8 & 16 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & -2 \\ 1 & 2 & 2 \\ -2 & 2 & -1 \end{bmatrix}$$

Next, we compute $A_1 = H_1A$:

$$A_1 = H_1A = \begin{bmatrix} -6 & 3 \\ 0 & 0 \\ 0 & 9 \end{bmatrix}.$$

Now we focus on the submatrix excluding the top row and left column of A_1 , which is $\widehat{A}_1 = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$. We

want to construct a reflector $\widehat{H}_2$ taking $u_2 = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$ to $v_2 = \begin{bmatrix} 9 \\ 0 \end{bmatrix}$. To do this, we set

$$p_2 = \frac{u_2 - v_2}{\|u_2 - v_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

and

$$\widehat{H}_2 = I - 2p_2p_2^T = I - \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then set $H_2 = \begin{bmatrix} 1 & 0 \\ 0 & \widehat{H}_2 \end{bmatrix}$. Multiplying H_2A_1 gives the matrix R in the QR factorization of A :

$$H_2H_1A = H_2A_1 = R = \begin{bmatrix} -6 & 3 \\ 0 & 9 \\ 0 & 0 \end{bmatrix}.$$

Inverting H_2H_1 , and using the fact that Householder reflectors are self-inverse, we obtain the Q factor from $A = H_1H_2R$, so that $Q = H_1H_2$, or

$$Q = \frac{1}{3} \begin{bmatrix} 2 & 1 & -2 \\ 1 & 2 & 2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ -2 & -1 & 2 \end{bmatrix},$$

which is the same as the one obtained in 1. (b) up to different choices of signs.